Instructions for Project

1. Application-Based Project

The objective of an application-based project is to understand an existing security concept, tool, technique, or solution, implement it, identify its limitations, and propose an improvement to address one of the identified challenges.

General Instructions

1. The topics provided are **broad areas**. Students are free to select a specific problem or application within the given topic.
2. The **same broad topic can be selected by more than one group**. However, each group must use a **different approach, methodology, implementation, tool, dataset, or application scenario**.
3. The project should not be limited to a theoretical study. Students are expected to **implement and demonstrate** the selected application/tool/technique.
4. Students should identify at least **one significant challenge or limitation** in the existing approach and propose an improvement to address it.
5. The proposed improvement should also be **implemented and evaluated** wherever feasible.
6. The project should include appropriate **experimental results, observations, comparisons, graphs/tables, or performance measurements**.

Report Structure

The report should contain the following sections:

1. **Introduction**
   - Introduction to the selected topic
   - Background and motivation
   - Importance of the topic in cyber security
   - Problem statement and objectives
2. **Overview of the Existing Article/Tool/Technique**
   - Detailed explanation of the selected article, tool, algorithm, framework, or technique
   - Working methodology
   - Architecture/workflow, if applicable
   - Advantages and applications
3. **Implementation of the Existing Approach**
   - Tools, technologies, programming languages, and datasets used
   - Implementation methodology
   - Experimental setup
   - Screenshots/results of the implementation
4. **Challenges and Limitations**
   - Identify the limitations or challenges of the existing approach
   - Explain why these challenges are important
   - Discuss their impact on security, performance, usability, scalability, etc.
5. **Proposed Approach**
   - Select **at least one challenge** identified in the previous section

- Propose a suitable solution or improvement
- Explain the proposed methodology
- Provide architecture/flowchart/pseudocode, wherever applicable

6. **Implementation of the Proposed Approach**
   - Implement the proposed solution
   - Describe the implementation process
   - Explain the experimental setup
7. **Results and Analysis**
   - Present the results obtained
   - Compare the existing and proposed approaches
   - Use tables, graphs, performance metrics, or other suitable evaluation methods
   - Discuss whether the proposed approach successfully addresses the identified challenge
8. **Conclusion**
   - Summarize the work carried out
   - Highlight the major findings
   - Discuss the effectiveness of the proposed approach
9. **References**
   - Include all research papers, articles, websites, tools, datasets, and other sources used.

2. Research-Based Project

The objective of a research-based project is to study existing research work in a selected security area, conduct a systematic survey and analysis of relevant research articles, identify research gaps or challenges, and propose possible approaches to address them.

General Instructions

1. The topics provided are **broad research areas**. Students are expected to narrow the selected topic into a specific research problem.
2. The **same topic can be selected by multiple groups**, but each group must focus on a **different research problem, set of papers, methodology, application area, or perspective**. 3. Students should study and critically analyse **multiple relevant research papers/articles** rather than relying on a single paper.
4. The survey should focus on identifying:
   - Existing approaches
   - Methodologies used
   - Datasets/tools used
   - Advantages and limitations
   - Performance/results
   - Research gaps and challenges
5. Students should not merely summarize the papers. They are expected to **compare, analyse, and critically evaluate** the existing research.
6. Based on the identified research gaps/challenges, students should suggest **one or more possible approaches** to address them.
7. Wherever feasible, students are encouraged to provide a **conceptual framework,**

architecture, algorithm, workflow, prototype, or preliminary implementation of their proposed approach.

Report Structure

1. **Introduction**
    - Introduction to the selected research area
    - Background and motivation
    - Importance of the topic in cyber security
    - Problem statement
    - Objectives of the study
2. **Literature Review**
    - Identify relevant research papers/articles
    - Provide an overview of each selected paper
    - Explain the problem addressed by each paper
    - Discuss the methodology/technique used
    - Summarize the major findings
3. **Comparative Survey of Existing Research**
    - Compare the selected research papers
    - Compare methodologies, datasets, tools, evaluation metrics, and results
    - Present the comparison using appropriate tables/figures
4. **Critical Analysis**
    - Analyse the strengths and weaknesses of existing approaches
    - Identify common limitations
    - Identify differences between the approaches
    - Discuss their applicability and effectiveness
5. **Research Challenges and Gaps**
    - Identify important unresolved challenges
    - Highlight research gaps found from the literature survey
    - Explain why these challenges/gaps require further investigation
6. **Proposed Approach**
    - Select one or more identified challenges/research gaps
    - Suggest a possible approach to address them
    - Explain the proposed methodology
    - Provide architecture, framework, workflow, algorithm, or conceptual model wherever appropriate
7. **Expected Outcomes / Preliminary Evaluation**
    - Discuss how the proposed approach could address the identified challenge
    - Define suitable evaluation parameters or metrics
    - If implemented, present preliminary results
    - Discuss expected advantages and limitations
8. **Conclusion and Future Scope**
    - Summarize the findings from the literature survey
    - Highlight the major research gaps identified
    - Summarize the proposed approach
    - Mention possible future research directions
9. **References**
    - Include all research papers, articles, books, datasets, websites, and other sources consulted.